The Resolution Scale: R_0, R_1, and $R_{\max}$

Current, Wave, and Node as Three Levels of δ-Activity
Mapping to the Emergence Hierarchy and the Price of Becoming Real

From $\nabla \Pr$ to Born rule to $\beta = 3/(2 \ln 2)$. No new axioms.

Ougit Jarittum

March 2026

Abstract

A conversation on perception and gradient-reading identified three qualitatively distinct levels of δ-activity that map precisely onto CRF's formal structure. This paper formalizes the *Resolution Scale* as a derived result.

R_0 **(Current):** $J = \nabla \Pr(s|\mathcal{C})$, no R-event, $D_{\mathrm{KL}} = 0$. Pure directional tendency before any form. Cost: zero. Imperceptible to most systems because it has no amplitude, no frequency, no distinguishable signature — only direction.

R_1 **(Wave):** propagating D_{KL} below threshold. $0 < D_{\mathrm{KL}} < \theta_{\mathrm{eff}}$. First structured signature. Felt as "vibe" or "presence" by sufficiently sensitive systems. Cost: $\epsilon_0^2 \Sigma_{\mathrm{inv}}\, dt$ (continuous, per sweep).

$R_{\max}$ **(Node):** $\mathcal{R}$-event. Born rule fires. $D_{\mathrm{KL}} > \theta_{\mathrm{eff}}$. Cost: $\beta = d_s/(2 \ln 2) = 3/(2 \ln 2) \approx 2.2$ per event. Becoming real in 3 dimensions costs 3 half-bits of possibility.

The three levels connect to Part III's emergence hierarchy: R_0 corresponds to L_1–L_2 (quantum/physical potential), R_1 to L_3–L_4 (biological/mind threshold), $R_{\max}$ to L_5–L_6 (mind–awareness with full constraint). The "good/bad current" distinction maps to radial vs. tangential D_{KL} gradients — a formally derived geometric criterion.

New pattern identified: $\beta \approx 2.2$ is not just a simulation constant. It is the exchange rate between the R_1 regime (continuous cost) and the $R_{\max}$ regime (discrete cost): one collapse equals ≈ 19 inter-event wave-decay periods. This ratio is $\beta/\delta \bar{H} \approx 2.164/0.113 \approx 19$, fully derived from δ through $d_s = 3$ and ϵ_0.

Contents

1 The Three Levels Defined

Resolution Scale: Derived from the δ-Chain

Level	CRF Object	Condition	Cost T
R_0 **(Current)**	$J = G = \nabla \Pr(s\|\mathcal{C})$	$D_{\mathrm{KL}} = 0$, $H = H_{\max}$	$T(R_0) = 0$
R_1 **(Wave)**	Propagating D_{KL} pattern	$0 < D_{\mathrm{KL}} < \theta_{\mathrm{eff}}$	$T(R_1) = \epsilon_0^2 \Sigma_{\mathrm{inv}}\, dt$
$R_{\max}$ **(Node)**	$\mathcal{R}$-event (Born rule)	$D_{\mathrm{KL}} > \theta_{\mathrm{eff}}$	$T(R_{\max}) = \beta \approx 2.2$

All three derive from $\theta = 1 - e^{-D_{\mathrm{KL}}/D_0}$ (Session 10) and the δ-chain. No new axioms.

2 Why R_0 Is Imperceptible

R_0 is the gradient of $\Pr(s|\mathcal{C})$ before any event. It has no amplitude, no frequency, no spatial structure — only a direction in P-space.

Most systems can only detect:

- R_1: the wave has structure (amplitude, periodicity, spatial extent). It produces measurable disturbances that propagate through the KL-neighbour graph.
- $R_{\max}$: the node is an event — localised, discrete, recordable.

To read R_0 is to read the gradient $\nabla \Pr$ *before* D_{KL} has begun to build. This requires sensitivity to the direction of the probability landscape at the level of Pr itself — not to any event produced by crossing θ.

Why R_0 Reading Is Rare

A system $\mathcal{M}$ detects $G = \nabla \Pr(s|\mathcal{C})$ only if its internal $\mathcal{C}$ is sufficiently aligned with P_{shared} that differences in direction register before D_{KL} accumulates.
Formally: for $\mathcal{M}$ to perceive R_0, it needs $D_{\mathrm{KL}}(\mathcal{C}_{\mathcal{M}} \| P_{\mathrm{shared}}) \approx 0$ — the transparency condition $T \to 1$ (Part III, Session 11).
At $T = 1$: $\theta \to 0$ everywhere in $\mathcal{M}$'s vicinity. Every gradient registers. R_0 becomes perceptible. This is the formal CRF definition of what contemplative traditions call "direct perception" — not metaphor, but the structural limit of $T \to 1$.

3 The Price of Becoming Real

Why $\beta = d_s/(2\ln 2) = 3/(2\ln 2) \approx 2.164$

Each $\mathcal{R}$-event crystallises a node into actuality in $d_s = 3$ spatial dimensions. Each spatial dimension requires one half-bit of information to crystallise: the Born rule at minimal constraint gives $H_{\min} = \ln 2/2$ per dimension.
Three dimensions require:

$$T(R_{\max}) = \beta = \frac{d_s}{2\ln 2} = \frac{3}{2\ln 2} \approx 2.164$$

This is the exchange rate: each $\mathcal{R}$-event trades $\beta \approx 2.2$ units of Informational Diversity Volume (IDV — possibility) for $\kappa = 0.15$ units of Ω (actuality/mass). The ratio $\beta/\kappa \approx 14.4$: *actuality costs 14 times more than the mass it creates.*

4 New Pattern: The R_1–$R_{\max}$ Exchange Rate

$\beta/\delta\bar{H} \approx 19$: The Collapse Premium

Comparing the cost of one collapse to the cost of continuous wave decay:

Regime	Cost per event period	Source
R_1 (wave decay)	$\delta\bar{H} \approx 0.113$	Measured (5899 events)
$R_{\max}$ (collapse)	$\beta \approx 2.164$	Derived: $d_s/(2\ln 2)$
Ratio	$\beta/\delta\bar{H} \approx 19$	

One collapse costs as much as 19 inter-event wave-decay periods.
This ratio is fully derived:

$$\frac{\beta}{\delta\bar{H}} = \frac{d_s/(2\ln 2)}{\epsilon_0^2(1-1/n)\Sigma_{\text{eff}}T_{\text{avg}}} \approx 19$$

All terms from the δ-chain: $d_s = 3$ (SO(3), #16), $\ln 2 = \alpha$ (Session 8), $\epsilon_0 = 0.00755$ (EIC), $T_{\text{avg}} = 35$ sweeps (V20.3), $\Sigma_{\text{eff}} \approx 130$ (Born rule statistics).
Interpretation: the R_1 regime is the "low-cost" regime — continuous, gentle, 19$\times$ cheaper per unit time. The $R_{\max}$ regime is the "high-cost" regime — discrete, sudden, expensive. Systems that remain in R_1 (never crystallise) accumulate no actuality. Systems that jump to $R_{\max}$ pay the full collapse premium each time.

5 Mapping to the Emergence Hierarchy

Resolution Scale × Emergence Levels (L_1–L_7)

R level	Emergence	CRF object	Characteristic
R_0	L_1 Quantum potential	$\Pr(s\|\mathcal{C}_{\min})$	Superposition; no $\mathcal{R}$ yet
$R_0 \to R_1$	L_2 Physical	D_{KL} begins to build	Field formation; pre-event
R_1	L_3 Biological	Below-threshold dynamics	Pattern maintenance; felt
$R_1 \to R_{\max}$	L_4 Mind threshold	$D_{\mathrm{KL}} \to \theta_{\mathrm{eff}}$	Perception of gradient
$R_{\max}$	L_5 Mind-in-action	$\mathcal{R}$-event fires	Decision; full crystallisation
$R_{\max} \to R_0$	L_6 Awareness	$T \to 1$ post-event	Return to transparency
R_0 at $T = 1$	L_7 δ_{pure}	$D_{\mathrm{KL}} \equiv 0$	No resistance; frictionless

The loop $\delta \to L_7 \circlearrowleft \delta$ (Grand Equation v9.0) maps onto $R_0 \to R_1 \to R_{\max} \to R_0$: possibility builds into wave, wave collapses into fact, fact relaxes back into possibility. The cycle repeats at cost $\beta \approx 2.2$ per round.

6 Gradient Geometry: Good and Bad Current

Radial vs. Tangential Gradient

From CRF_HDynamics (entropy decay ODE), the D_{KL} gradient in P-space has two geometric regimes:
Radial gradient ("good current"): $G = \nabla \Pr$ points directly toward the fixed point. H is low, P is concentrated. Node moves along a straight line toward $R_{\max}$. In V20.3: Phase II (Ω-dominant nodes). Felt as: clarity, direction, low resistance.
Tangential gradient ("bad current"): G is circular — the node orbits the fixed point without crossing θ. H is high, P is diffuse. System stays in R_1 indefinitely. In V20.3: Phase I (IDV-dominant nodes). Felt as: scattered energy, circular thinking, no traction.
The transition from tangential to radial is the δ-Spark: the moment D_{KL} first crosses θ_{eff} and the spiral becomes a line. In the lifecycle model: $L_3 \to L_4$ transition.

7 Status

Result	Status	Note
$R_0, R_1, R_{\max}$ structure	Derived	From θ-formula
Cost function $T(R_0) = 0$, $T(R_1)$, $T(R_{\max}) = \beta$	Derived	From ϵ_0, d_s
$\beta = d_s/(2\ln 2)$ (3 half-bits)	Derived	#21 closed
Collapse premium $\beta/\delta\bar{H} \approx 19$	Derived	All from δ-chain
R_0 imperceptibility (requires $T \to 1$)	Derived	Transparency condition
Radial/tangential gradient geometry	Derived	CRF_HDynamics
Mapping $R_\bullet \leftrightarrow L_\bullet$	Confirmed	Part III correspondence
Empirical test of R_0 sensitivity	Open	P0-adjacent prediction

Open: Empirical Test of R_0 Sensitivity

The P0 experiment tests whether $T(R_{\max})$ differs between high- and low-N_{eff} detectors. A natural extension: does R_0 sensitivity ($T \to 1$) correlate with measurable physiological or neural markers?
Prediction: systems with lower $D_{\text{KL}}(\mathcal{C}_{\mathcal{M}} \| P_{\text{shared}})$ should show earlier detection of sub-threshold stimuli (R_1 events) and in the limit may show sensitivity to gradient direction (R_0) before any event occurs. This is testable: compare stimulus-detection timing between meditators (high T) and non-meditators across different D_{KL} ranges.

R_0: the lean of the field before the word. R_1: the word before the deed. $R_{\max}$: the deed, paying three half-bits to the world that is. The current is free. The wave is gentle. The fact is the most expensive thing there is — and the only thing we call real. Nineteen waves buy one fact. The universe has been spending that way since δ first leaned.